# MGM Macau and Cotai

## Resource Efficiency Management Report

September 2021

Energenz Consulting Limited

# EXECUTIVE SUMMARY

Energenz has been working closely with MGM to implement the EMOs as well as identify additional opportunities to achieve ISO certification energy targets. Energenz appreciates the great support and trust from MGM team over the years, in particular the Facility Management and the Sustainability Teams.

## Key Performance Indicators

Energenz reports both normalized and actual savings on a monthly basis for both properties. The normalized savings are determined based on a 2019 baseline, with aligned annual savings targets to meet both ISO50001 as well as Energenz's retainer contract requirements. Please note that the reported normalized savings achieved are to-date and cumulative for each month of the respective year.

**Table 1. ISO and Retainer Contract Energy Performance Targets for 2025**

| MGM Properties | Baseline | Energy Savings | Normalized Savings Achieved Up to September 2021 | 2022 | 2023 | 2024 | 2025 |
|---|---|---|---|---|---|---|---|
| **Macau** | 2019 | Target | -4% | -8% | -12% | N/A | **-13%** |
| | | Actual | **-7.1%** | - | - | - | - |
| **Cotai** | 2019 | Target | -4% | -8% | -12% | N/A | **-13%** |
| | | Actual | **-6.1%** | - | - | - | - |

**MGM Cotai ISO energy performance: e-bus consumption is included.*

The actual monthly savings are determined against the 2019 baseline consumption and cumulative for each month of 2021.

**Table 2. Internal Actual Energy Performance Targets for 2025**

| MGM Properties | Baseline | Actual Saving Target[1] | Savings Achieved Up to September 2021 |
|---|---|---|---|
| **Macau** | 2019 | 15% | **-7.2%** |
| **Cotai** | 2019 | 15% | **-4.4%** |

## Highlighted Changes in Operation

MGM Cotai has fully utilized Heat Pump for maintaining DHW temperatures and disinfecting the hot water system from 6 August 2021.

Due to high temperature alarm of DRUPS equipment, MGM Cotai has installed 2 split type AC for reducing the intake air temperature. This practice started on 1 August 2021 and ended on 30 September 2021.

Steam to water heat exchanger in MGM Macau has been switched on for dehumidification at low zone guest room (L5 – L17) starting from 14 July 2021. This practice is continued in September 2021.

[1] Actual Energy Savings Target for 2025 has been established based on future CAPEX projects, following review with MGM teams

### Consumption Intensity

The YTD electricity and water use intensities are tracked on a monthly basis for both properties:

**Table 3. Year to Date Actual Electricity Use Intensity (EUI)**

| MGM Properties | YTD EUI (kWh/$m^2$) | % Change Compared to 2019 |
|---|---|---|
| **Macau** | 26.2 | **-8.2%** |
| **Cotai** | 27.2 | **-2.9%** |

**Table 4. Year to Date Actual Water Use Intensity (WUI)**

| MGM Properties | YTD WUI ($m^3$/ $m^2$) | % Change Compared to 2019 |
|---|---|---|
| **Macau** | 0.21 | **-12.5%** |
| **Cotai** | 0.17 | **-14.4%** |

*Note: Cotai Area: 374,138 $m^2$ Macau Area: 211,545 $m^2$*

## ISO 50001 Certification Audit

### Macau

The ISO 50001 certificate has been received by MGM on 1 April 2021. The certificate will be valid from 3 April 2021 to 2 April 2024.

### Cotai

The next external surveillance audit will be scheduled for 22 October 2021.

## Target Milestones for Year 2021

The following activities are planned for completion this year:

- To perform ISO 50001 2021 Internal and External Audit for MGM Macau and MGM Cotai
- To deploy and fine tune DGLux dashboard with heatmap development extension
- To complete analysis on MGM Cotai spectacle thermal comfort investigation
- To track and monitor changes in air change rates per hour for regular reporting
- To investigate and implement the below list of projects during Year 2021

**Table 5. MGM Macau Year 2021 Projects**

| Item | Initiatives |
|---|---|
| **Electricity** | |
| 1 | Smoking Lounge Exhaust Fans Operation Optimization |
| 2 | Optimize Winter Operation of Cooling Tower Control Strategy - Fan Control |
| 3 | Air Change Rate (ACH) Optimization |
| 4 | Implement Motor Efficiency Upgrades (24 units) |
| 5 | Implement Static Pressure Set-Point Reset in AHU (10 units) |
| 6 | Chiller Plant Optimization: Chilled Water Differential Pressure Set-point Optimization |
| 7 | Ball Room VAV Recommissioning/Rectification Work |
| 8 | Las Vegas Room (LVR) Gaming Area VAV Recommissioning/Rectification Work |
| 9.1 | PICCV Implementation – Ball room/Foyer AHU (3 Units) |
| 9.2 | PICCV Implementation – RF AHU (2 Units) |
| 9.3 | PICCV Implementation (17 Units) |
| 10 | Neon Lighting LED Replacement |
| **Water** | |
| 11 | Optimize Existing Water Softener (Podium) |

**Table 6. MGM Cotai Year 2021 Projects**

| Item | Initiatives |
|---|---|
| **Electricity** | |
| 1 | Air Change Rate Optimization |
| 2 | B1 Carpark and Loading Bay Lighting Operation Schedule Optimization |
| 3 | Kitchen Exhaust Fan (KEF) Demand Control Ventilation (DCV) and Hood Control for Remaining Kitchen's (e.g. Banquet, Chun, Mansion, Coast.) |
| 4 | Optimize BOH FCU and VAV Temperature Set-Point |
| 5 | Optimize Car Park Ventilation Fans Control |
| 6 | Optimize Hotel Tower Circulation Pumps Operation |
| 7 | Chiller Plant Optimization:<br>1. Review Chillers Staging<br>2. Cooling Tower Fan Speed Optimization<br>3. Condensing Water Pump Speed Optimization |
| 8 | Chilled Water Differential Pressure Set-Point Reset Logic Implementation |
| 9 | B2 Domestic Hot Water Heat Exchanger Upgrade for fully utilizing the heat pump |
| 10 | Implement Static Pressure & Supply Air Temperature Set-Point Reset in AHU (48 Units) |
| 11 | PICCV Implementation (7 Units) |
| 12 | Solar PV Panel Implementation |
| **Water** | |
| 13 | Extension of CDP from ST-CD-02 to ST-CD-01 for Cooling Tower Makeup |

# TABLE OF CONTENT


EXECUTIVE SUMMARY 2
ISO 50001 Certification Audit 3
Target Milestones for Year 2021 4
1. Portfolio Performance Summary 7
1.1 Electricity Use Intensity (EUI) 7
1.2 Water Use Intensity (WUI) 8
2. MGM Macau Performance 9
2.1 Electricity Consumption 9
2.2 LPG Consumption 11
2.3 ISO50001 Energy Consumption Record 13
2.4 Water Consumption 14
2.5 Macau Year 2021 Projects 16
3. MGM Cotai Performance 17
3.1 Electricity 17
3.2 Natural Gas Consumption 20
3.3 ISO50001 Energy Consumption Record 22
3.4 Water Consumption 23
3.5 Cotai Year 2021 Projects 25
Appendix A 26
Regression Analysis 26
MGM Macau 26
MGM Cotai 26

# ACKNOWLEDGEMENT

Energenz has been working with MGM Team as the energy manager to achieve energy savings in MGM properties and feed operational intelligence into the design.

The major goal of this project is not only to realize operation energy savings but also extending the life of equipment, hence reduce CAPEX. MGM has demonstrated continuous commitment to be an environmentally and socially responsible business in the past years. In addition to that Energenz has assisted MGM Macau and Cotai to achieve several other energy management and sustainability awards and certifications, such as ISO 50001 and CEM Energy Savings Award.

Energenz appreciates the great support and trust from MGM Team over the years, especially the Facilities Management and the Sustainability Teams.

# 1. Portfolio Performance Summary

## 1.1 Electricity Use Intensity (EUI)

Macau's electricity uses intensity (EUI) in September 2021 had increased by **2.5%** compared to same month in 2019 and decreased by **8.2%** in 2021 compared to 2019. Meanwhile, Cotai's EUI in September 2021 has increased by **4.1%** compared to same month in 2019 and decreased by **2.9%** in 2021 compared to 2019.

**Table 7. Actual Electricity Consumption Intensity (EUI) Comparison**

| Monthly Period | Macau | | | Cotai | | |
|---|---|---|---|---|---|---|
| | Year 2019 Actual EUI (kWh/m²) | Year 2021 Actual EUI (kWh/m²) | % Change | Year 2019 Actual EUI (kWh/m²) | Year 2021 Actual EUI (kWh/m²) | % Change |
| **January** | 24.4 | 20.4 | -16.3% | 24.5 | 21.8 | -10.7% |
| **February** | 23.0 | 19.8 | -14.0% | 22.9 | 20.4 | -11.0% |
| **March** | 25.4 | 23.3 | -8.4% | 25.0 | 24.6 | -1.6% |
| **April** | 27.4 | 23.8 | -13.0% | 27.3 | 25.4 | -6.8% |
| **May** | 30.0 | 28.9 | -3.6% | 29.2 | 30.6 | +4.6% |
| **June** | 32.1 | 28.7 | -10.4% | 31.0 | 29.7 | -4.2% |
| **July** | 33.0 | 30.8 | -6.8% | 32.0 | 31.7 | -1.0% |
| **August** | 32.1 | 29.8 | -7.2% | 31.2 | 30.5 | -2.3% |
| **September** | 29.6 | 30.3 | +2.5% | 28.8 | 30.0 | +4.1% |
| **October** | | | | | | |
| **November** | | | | | | |
| **December** | | | | | | |
| **Average** | **28.6** | **26.2** | **-8.2%** | **28.0** | **27.2** | **-2.9%** |

## 1.2 Water Use Intensity (WUI)

Macau's water uses intensity (WUI) in September 2021 decreased by **3.9%** compared to the same month in 2019 and decreased by **12.5 %** in 2021 compared to 2019. Meanwhile, Cotai's WUI in September 2021 decreased by **2.6%** compared to the same month in 2019 and decreased by **14.4%** in 2021 compared to 2019.

**Table 8. Actual Water Consumption Intensity (WUI) Comparison**

| Monthly Period | Macau | | | Cotai | | |
|---|---|---|---|---|---|---|
| | Year 2019 Actual WUI ($m^3/m^2$) | Year 2021 Actual WUI ($m^3/m^2$) | % Change | Year 2019 Actual WUI ($m^3/m^2$) | Year 2021 Actual WUI ($m^3/m^2$) | % Change |
| **January** | 0.22 | 0.19 | -13.2% | 0.16 | 0.12 | -26.2% |
| **February** | 0.22 | 0.19 | -11.9% | 0.17 | 0.14 | -22.0% |
| **March** | 0.21 | 0.20 | -2.5% | 0.17 | 0.15 | -11.8% |
| **April** | 0.24 | 0.24 | -2.6% | 0.20 | 0.18 | -10.9% |
| **May** | 0.24 | 0.25 | +2.6% | 0.20 | 0.21 | +5.7% |
| **June** | 0.27 | 0.21 | -22.2% | 0.21 | 0.18 | -16.6% |
| **July** | 0.28 | 0.22 | -23.1% | 0.23 | 0.19 | -18.6% |
| **August** | 0.28 | 0.20 | -28.9% | 0.23 | 0.17 | -26.3% |
| **September** | 0.24 | 0.23 | -3.9% | 0.19 | 0.19 | -2.6% |
| **October** | | | | | | |
| **November** | | | | | | |
| **December** | | | | | | |
| **Average** | **0.25** | **0.21** | **-12.5%** | **0.20** | **0.17** | **-14.4%** |

# 2. MGM Macau Performance

## 2.1 Electricity Consumption

The actual electricity consumption and cost in September 2021 increased by **2.5%** and **3.4%** respectively compared to same month in 2019. To date, MGM Macau has decreased the actual electricity consumption and associated costs by **8.2%** and **9.5%** respectively.

**Table 9. Macau - Summary of Actual Monthly Electricity Consumption and Cost**

| Monthly Period | Actual Electricity Consumption (kWh) | | | Actual Electricity Cost (MOP) | | | % Change – CDD |
|---|---|---|---|---|---|---|---|
| | 2019 | 2021 | % Change - Actual | 2019 | 2021 | % Change - Actual Cost | |
| **January** | 5,151,480 | 4,310,160 | -16.3% | 6,038,725 | 4,717,260 | -21.9% | -18.5% |
| **February** | 4,870,560 | 4,190,760 | -14.0% | 5,728,930 | 4,591,170 | -19.9% | -3.8% |
| **March** | 5,381,640 | 4,931,880 | -8.4% | 6,311,830 | 5,387,480 | -14.6% | +39.0% |
| **April** | 5,796,240 | 5,042,160 | -13.0% | 6,681,548 | 5,808,340 | -13.1% | -5.5% |
| **May** | 6,349,920 | 6,122,640 | -3.6% | 7,312,310 | 7,034,530 | -3.8% | +43.8% |
| **June** | 6,780,240 | 6,072,960 | -10.4% | 8,582,460 | 7,681,070 | -10.5% | -4.4% |
| **July** | 6,984,480 | 6,506,760 | -6.8% | 8,752,550 | 8,217,570 | -6.1% | +1.4% |
| **August** | 6,799,320 | 6,306,840 | -7.2% | 8,526,375 | 7,960,170 | -6.6% | -4.7% |
| **September** | 6,263,400 | 6,418,920 | +2.5% | 7,853,170 | 8,118,840 | +3.4% | +7.9% |
| **October** | | | | | | | |
| **November** | | | | | | | |
| **December** | | | | | | | |
| **Total/Average** | **54,377,280** | **49,903,080** | **-8.2%** | **65,787,897** | **59,516,430** | **-9.5%** | **+5.9%** |

***Important note:*** *Cooling Degree Days (CDDs) is the main variable taken into consideration on Electricity consumption normalization.*

Energenz summarized the key items which could result in electricity consumption increase below:

- The chiller plant consumption has increased by 15.5% in September 2021 because CDD has been increased by 7.9% compared to same month in 2019.

Projects on-going/completed which contributed to electricity consumption decrease below:

- Outdoor billboards lighting equipment upgrades
- Implement PICCV in AHUs (17 units)
- Implement Static Pressure Set-Point Reset in AHUs (10 units)
- Implement Motor Efficiency Upgrades (24 units)
    - Total 9 units was completed till September 2021

Table 10 presents the comparison of electricity rates in the year of 2019 and 2021. The electricity rate in September 2021 increased by **0.9%** compared to same month in 2019.

**Table 10. Macau - Summary of Actual Monthly Electricity Rate**

| Monthly Period | Year 2019 – Electricity Rate (MOP/kWh) | Year 2021 – Electricity Rate (MOP/kWh) | % Change – Electricity Rate |
|---|---|---|---|
| **January** | 1.17 | 1.09 | -6.6% |
| **February** | 1.18 | 1.10 | -6.9% |
| **March** | 1.17 | 1.09 | -6.9% |
| **April** | 1.15 | 1.15 | -0.1% |
| **May** | 1.15 | 1.15 | -0.2% |
| **June** | 1.27 | 1.26 | -0.1% |
| **July** | 1.25 | 1.26 | +0.8% |
| **August** | 1.25 | 1.26 | +0.6% |
| **September** | 1.25 | 1.26 | +0.9% |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **1.21** | **1.18** | **-2.0%** |

Table 11 summarizes the monthly normalized electricity consumption. The normalized electricity consumption in September 2021 decreased by **4.7%** compared to same month in 2019.

**Table 11. Macau - Summary of Normalized Monthly Electricity Consumption**

| Monthly Period | 2019 Normalized Electricity Consumption (kWh) | 2021 Actual Electricity Consumption (kWh) | % Change - Normalized |
|---|---|---|---|
| **January** | 4,763,370 | 4,310,160 | -9.5% |
| **February** | 5,024,953 | 4,190,760 | -16.6% |
| **March** | 5,398,600 | 4,931,880 | -8.6% |
| **April** | 5,765,334 | 5,042,160 | -12.5% |
| **May** | 6,578,889 | 6,122,640 | -6.9% |
| **June** | 6,488,142 | 6,072,960 | -6.4% |
| **July** | 6,770,179 | 6,506,760 | -3.9% |
| **August** | 6,594,158 | 6,306,840 | -4.4% |
| **September** | 6,734,744 | 6,418,920 | -4.7% |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **54,118,370** | **49,903,080** | **-7.8%** |

## 2.2 LPG Consumption

LPG consumption can be broken down into three (3) categories: boiler, hotel upper zone DHW and kitchens/restaurants. Since the boiler and restaurants LPG utilization cover over 95% of the overall site LPG consumption, Energenz focuses the analysis on those two areas.

According to Table 12, the actual LPG consumption and cost in September 2021 decreased by **1.3%** and increased by **45.1%** respectively compared to same month in 2019. To date MGM Macau has decreased the actual LPG consumption by **0.8%** and associated costs increased by **23.4%**.

**Table 12. Macau - Summary of Actual Monthly LPG Consumption and Cost**

| Monthly Period | Actual LPG Consumption (kg) | | | Actual LPG Cost (MOP) | | | % Change – Min OAT |
|---|---|---|---|---|---|---|---|
| | 2019 | 2021 | % Change - Actual | 2019 | 2021 | % Change - Actual Cost | |
| **January** | 94,330 | 121,398 | +28.7% | 706,171 | 947,708 | +34.2% | -29.9% |
| **February** | 72,191 | 74,797 | +3.6% | 415,818 | 632,778 | +52.2% | +10.1% |
| **March** | 78,807 | 84,131 | +6.8% | 453,930 | 711,751 | +56.8% | +6.4% |
| **April** | 81,430 | 77,702 | -4.6% | 522,530 | 652,357 | +24.8% | -2.5% |
| **May** | 84,021 | 71,294 | -15.1% | 644,444 | 578,906 | -10.2% | +20.1% |
| **June** | 78,030 | 62,104 | -20.4% | 598,488 | 504,285 | -15.7% | -4.6% |
| **July** | 73,706 | 71,398 | -3.1% | 533,791 | 579,753 | +8.6% | +0.4% |
| **August** | 79,296 | 74,204 | -6.4% | 507,497 | 698,258 | +37.6% | -1.2% |
| **September** | 75,413 | 74,426 | -1.3% | 482,646 | 700,353 | +45.1% | +5.0% |
| **October** | | | | | | | |
| **November** | | | | | | | |
| **December** | | | | | | | |
| **Total/Average** | **717,226** | **711,454** | **-0.8%** | **4,865,314** | **6,006,149** | **+23.4%** | **+1.4%** |

***Important note:*** *Minimum Outdoor Air Temperature (OAT) is the main variable taken into consideration on LPG consumption normalization.*

Energenz summarized the key items which could result in LPG consumption decrease below:

- Boilers LPG consumption increased by 10.6% in September 2021 because steam to water heat exchanger has been switched on for dehumidification at low zone guest room (L5 – L17) from 14 July 2021.
- LPG consumption from Kitchen and Restaurants decreased by 5.7% in September 2021 compared with the same month in 2019.

Table 13 presents the comparison of LPG rates in the year of 2019 and 2021. The LPG rate in September 2021 increased by **47.0%** compared with the same month in 2019.

**Table 13. Macau - Summary of Actual Monthly LPG Rate**

| Monthly Period | Year 2019 – LPG Rate (MOP/kg) | Year 2021 – LPG Rate (MOP/ kg) | % Change – LPG Rate |
|---|---|---|---|
| **January** | 7.49 | 7.81 | +4.3% |
| **February** | 5.76 | 8.46 | +46.9% |
| **March** | 5.76 | 8.46 | +46.9% |
| **April** | 6.42 | 8.40 | +30.8% |
| **May** | 7.67 | 8.12 | +5.9% |
| **June** | 7.67 | 8.12 | +5.9% |
| **July** | 7.24 | 8.12 | +12.1% |
| **August** | 6.40 | 9.41 | +47.0% |
| **September** | 6.40 | 9.41 | +47.0% |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **6.76** | **8.48** | **+25.5%** |

Table 14 summarizes the monthly normalized LPG consumption. The normalized consumption in September 2021 decreased by **6.0%** compared with the actual consumption in 2021.

**Table 14. Macau - Summary of Normalized Monthly LPG Consumption**

| Monthly Period | 2019 Normalized LPG Consumption (kg) | 2021 Actual LPG Consumption (kg) | % Change – Normalized |
|---|---|---|---|
| **January** | 116,973 | 121,398 | +3.8% |
| **February** | 83,731 | 74,797 | -10.7% |
| **March** | 81,839 | 84,131 | +2.8% |
| **April** | 71,748 | 77,702 | +8.3% |
| **May** | 74,993 | 71,294 | -4.9% |
| **June** | 72,822 | 62,104 | -14.7% |
| **July** | 77,363 | 71,398 | -7.7% |
| **August** | 74,397 | 74,204 | -0.3% |
| **September** | 79,156 | 74,426 | -6.0% |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **733,021** | **711,454** | **-2.9%** |

## 2.3 ISO50001 Energy Consumption Record

The Macau ISO50001 energy performance target is to achieve 13% normalized energy savings by 2025 compared to 2019 baseline. Table 15 below summarizes the site monthly energy consumption (Electricity and LPG combined) compared to 2019 baseline. To date, normalized energy saving achieved up to September 2021 is **7.1%**.

**Table 15. Macau ISO 50001 Normalized Monthly Energy Consumption**

| Monthly Period | 2019 Normalized Energy Consumption (GJ) | 2021 Actual Energy Consumption (GJ) | % Change - Normalized (2021 Vs 2019) |
|---|---|---|---|
| **January** | 22,447 | 21,016 | -6.4% |
| **February** | 21,883 | 18,475 | -15.6% |
| **March** | 23,142 | 21,566 | -6.8% |
| **April** | 24,005 | 21,672 | -9.7% |
| **May** | 27,081 | 25,271 | -6.7% |
| **June** | 26,656 | 24,676 | -7.4% |
| **July** | 27,877 | 26,659 | -4.4% |
| **August** | 27,109 | 26,066 | -3.8% |
| **September** | 27,831 | 26,480 | -4.9% |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **228,032** | **211,880** | **-7.1%** |

***Important note:*** *Factors used for fuels' unit conversion: 1) Electricity (0.0036) and LPG (0.0453)*

## 2.4 Water Consumption

Based on the existing water metering system, the overall site water consumption can be broken down into four (4) categories: Kitchen and Restaurants; Hotel/Tower; Fire Services; and Chiller Plant & miscellaneous[2].

According to Table 16, the actual water consumption in September 2021 decreased by **3.9%** compared to same month in 2019. To date, MGM Macau has decreased actual water consumption and associated costs by **12.5%** and **12.8%** respectively.

**Table 16. Macau - Summary of Actual Monthly Water Consumption and Cost**

| Monthly Period | Actual Water Consumption ($m^3$) | | | Actual Water Cost (MOP) | | | Normalization Factors | |
|---|---|---|---|---|---|---|---|---|
| | 2019 | 2021 | % Change Actual | 2019 | 2021 | % Change Actual Cost | % Change – CDD | % Change – Food Covers |
| **January** | 46,006 | 39,952 | -13.2% | 356,970 | 299,330 | -16.1% | -18.5% | -55.0% |
| **February** | 46,637 | 41,079 | -11.9% | 361,510 | 317,780 | -12.1% | -3.8% | -47.7% |
| **March** | 44,276 | 43,168 | -2.5% | 342,840 | 334,000 | -2.6% | +39.0% | -36.4% |
| **April** | 51,156 | 49,819 | -2.6% | 402,280 | 386,600 | -3.9% | -5.5% | -35.8% |
| **May** | 51,800 | 53,145 | +2.6% | 401,890 | 420,070 | +4.5% | +43.8% | -35.9% |
| **June** | 57,243 | 44,508 | -22.2% | 444,090 | 345,320 | -22.2% | -4.4% | -44.0% |
| **July** | 60,099 | 46,227 | -23.1% | 466,230 | 358,640 | -23.1% | +1.4% | -32.0% |
| **August** | 60,050 | 42,681 | -28.9% | 465,790 | 331,220 | -28.9% | -4.7% | -54.4% |
| **September** | 50,479 | 48,499 | -3.9% | 391,440 | 376,290 | -3.9% | +7.9% | -26.8% |
| **October** | | | | | | | | |
| **November** | | | | | | | | |
| **December** | | | | | | | | |
| **Total/Average** | **467,746** | **409,078** | **-12.5%** | **3,633,040** | **3,169,250** | **-12.8%** | **+5.9%** | **-41.4%** |

***Important note:*** *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

Energenz summarized the key items which could result in water consumption decrease below:

- Hotel Tower water consumption decreased by 15.5% compared to same month in 2019. Moreover, the guest occupancy decreased by 22.9% compared to same month in 2019.
- Podium water consumption for September 2021 increased by 0.6% compared to the same month in 2019. Moreover, the CDD increased by 7.9% compared to the same month in 2019.

[2] Miscellaneous includes the water consumption from toilets, laundry, etc.

Table 17 presents the comparison of water rates in the year of 2019 and 2021. The water rate in September 2021 slightly increased by **0.1%** compared to same month in 2019. The average utility rate of water has slightly decreased by **0.3%** compared to 2019.

**Table 17. Macau - Summary of Actual Monthly Water Rates**

| Monthly Period | Year 2019 – Water Rate (MOP/m³) | Year 2021 – Water Rate (MOP/m³) | % Change – Water Rate |
|---|---|---|---|
| **January** | 7.76 | 7.49 | -3.4% |
| **February** | 7.75 | 7.74 | -0.2% |
| **March** | 7.74 | 7.74 | -0.1% |
| **April** | 7.86 | 7.76 | -1.3% |
| **May** | 7.76 | 7.90 | +1.9% |
| **June** | 7.76 | 7.76 | +0.01% |
| **July** | 7.76 | 7.76 | +0.01% |
| **August** | 7.76 | 7.76 | +0.05% |
| **September** | 7.75 | 7.76 | +0.1% |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **7.77** | **7.74** | **-0.3%** |

Table 18 summarized the monthly normalized water consumption. The normalized water consumption in September 2021 decreased by **0.1%** compared to the actual consumption in 2021.

**Table 18. Macau - Summary of Normalized Monthly Water Consumption**

| Monthly Period | 2019 Normalized Water Consumption (m³) | 2021 Actual Water Consumption (m³) | % Change - Normalized |
|---|---|---|---|
| **January** | 31,258 | 39,952 | +27.8% |
| **February** | 34,783 | 41,079 | +18.1% |
| **March** | 39,846 | 43,168 | +8.3% |
| **April** | 42,227 | 49,819 | +18.0% |
| **May** | 49,504 | 53,145 | +7.4% |
| **June** | 46,073 | 44,508 | -3.4% |
| **July** | 51,711 | 46,227 | -10.6% |
| **August** | 45,178 | 42,681 | -5.5% |
| **September** | 48,554 | 48,499 | -0.1% |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **389,073** | **409,078** | **+5.1%** |

## 2.5 Macau Year 2021 Projects

Energenz will perform analysis and coordinate with MGM to implement the following projects in 2021 and reduce the site energy and water consumption which supporting Energenz annual normalized energy savings target of 2%.

**Table 19. MGM Macau Year 2021 Projects**

| Item | Initiatives |
|---|---|
| **Electricity** | |
| 1 | Smoking Lounge Exhaust Fans Operation Optimization |
| 2 | Optimize Winter Operation of Cooling Tower Control Strategy - Fan Control |
| 3 | Air Change Rate (ACH) Optimization |
| 4 | Implement Motor Efficiency Upgrades (24 units) |
| 5 | Implement Static Pressure Set-Point Reset in AHU (10 units) |
| 6 | Chiller Plant Optimization: Chilled Water Differential Pressure Set-point Optimization |
| 7 | Ball Room VAV Recommissioning/Rectification Work |
| 8 | Las Vegas Room (LVR) Gaming Area VAV Recommissioning/Rectification Work |
| 9.1 | PICCV Implementation – Ball room/Foyer AHU (3 Units) |
| 9.2 | PICCV Implementation – RF AHU (2 Units) |
| 9.3 | PICCV Implementation (17 Units) |
| 10 | Neon Lighting LED Replacement |
| **Water** | |
| 11 | Optimize Existing Water Softener (Podium) |

# 3. MGM Cotai Performance

## 3.1 Electricity

The actual electricity consumption and cost in September 2021 increased by **4.1%** and **4.7%** respectively compared to the same month in 2019. To date, MGM Cotai has decreased actual electricity consumption and associated costs by **2.6%** and **4.2%** respectively.

**Table 20. Cotai - Summary of Actual Monthly Electricity Consumption and Cost**

| Monthly Period | Actual Electricity Consumption (kWh) | | | Actual Electricity Cost (MOP) | | | % Change – CDD |
|---|---|---|---|---|---|---|---|
| | 2019 | 2021 | % Change - Actual | 2019 | 2021 | % Change - Actual Cost | |
| **January** | 8,938,560 | 8,168,400 | -8.6% | 10,919,060 | 9,315,910 | -14.7% | -18.5% |
| **February** | 8,560,200 | 7,637,640 | -10.8% | 10,478,873 | 8,737,740 | -16.6% | -3.8% |
| **March** | 9,350,520 | 9,197,160 | -1.6% | 11,429,480 | 10,499,490 | -8.1% | +39.0% |
| **April** | 10,207,920 | 9,517,560 | -6.8% | 12,272,805 | 11,430,660 | -6.9% | -5.5% |
| **May** | 10,940,160 | 11,446,440 | +4.6% | 13,125,310 | 13,735,880 | +4.7% | +43.8% |
| **June** | 11,610,000 | 11,122,320 | -4.2% | 13,938,080 | 13,340,650 | -4.3% | -4.4% |
| **July** | 11,977,200 | 11,853,120 | -1.0% | 14,246,260 | 14,194,900 | -0.4% | +1.4% |
| **August** | 11,685,960 | 11,412,960 | -2.3% | 13,897,965 | 13,675,130 | -1.6% | -4.7% |
| **September** | 10,774,080 | 11,220,240 | +4.1% | 12,829,680 | 13,451,090 | +4.8% | +7.9% |
| **October** | | | | | | | |
| **November** | | | | | | | |
| **December** | | | | | | | |
| **Total/Average** | **94,044,600** | **91,575,840** | **-2.6%** | **113,137,513** | **108,381,450** | **-4.2%** | **+5.9%** |

***Important note**: Cooling Degree Days (CDDs) is the main variable taken into consideration on Electricity consumption normalization. The e-Bus electricity consumption has been included with the presented electricity consumption above.*

Energenz summarized the key items which could result in electricity consumption increase below:

- The chiller consumption has increased by 23.7% in September 2021 because CDD has been increased by 7.9% compared to same month in 2019.
- The heat pump consumption has increased by 14.9% in September 2021 because the utilization of Heat Pump for maintaining DHW temperatures and disinfecting the hot water system from 6 August 2021.

Table 21 presents the monthly e-bus electricity consumption in MGM Cotai. The e-bus electricity consumption in September 2021 decreased by **45.7%** compared to same month in 2019. Please note that consumption related to the e-bus generally increases as compared to 2019 as E-bus operation started in January 2019.

**Table 21. Cotai - Summary of Monthly e-bus Electricity Consumption**

| Monthly Period | E-bus Electricity Consumption (kWh) | | |
|---|---|---|---|
| | 2019 | 2021 | % Change - Actual |
| January | 28,305 | 44,839 | +58.4% |
| February | 24,007 | 32,534 | +35.5% |
| March | 34,315 | 34,902 | +1.7% |
| April | 36,147 | 37,412 | +3.5% |
| May | 37,240 | 51,584 | +38.5% |
| June | 54,293 | 56,629 | +4.3% |
| July | 66,199 | 64,410 | -2.7% |
| August | 74,208 | 44,185 | -40.5% |
| September | 74,828 | 40,618 | -45.7% |
| October | | | |
| November | | | |
| December | | | |
| Total/Average | **429,541** | **407,112** | **-5.2%** |

Table 22 presents the comparison of electricity rates in the year of 2019 and 2021. The electricity rate in September 2021 increased by **0.7%** compared to same month in 2019.

**Table 22. Cotai - Summary of Actual Monthly Electricity Rates**

| Monthly Period | Year 2019 – Electricity Rate (MOP/kWh) | Year 2021 – Electricity Rate (MOP/kWh) | % Change – Electricity Rate |
|---|---|---|---|
| January | 1.22 | 1.14 | -6.6% |
| February | 1.22 | 1.14 | -6.5% |
| March | 1.22 | 1.14 | -6.6% |
| April | 1.20 | 1.20 | -0.1% |
| May | 1.20 | 1.20 | +0.02% |
| June | 1.20 | 1.20 | -0.1% |
| July | 1.19 | 1.20 | +0.7% |
| August | 1.19 | 1.20 | +0.8% |
| September | 1.19 | 1.20 | +0.7% |
| October | | | |
| November | | | |
| December | | | |
| Total/Average | **1.20** | **1.18** | **-2.0%** |

Table 23 summarized the monthly normalized electricity consumption. The normalized electricity consumption in September 2021 decreased by **3.2%** compared to same month in 2019.

**Table 23. Cotai - Summary of Normalized Monthly Electricity Consumption**

| Monthly Period | 2019 Normalized Electricity Consumption (kWh) | 2021 Actual Electricity Consumption (kWh) | % Change - Normalized |
|---|---|---|---|
| **January** | 8,594,286 | 8,168,400 | -5.0% |
| **February** | 8,991,397 | 7,637,640 | -15.1% |
| **March** | 9,558,635 | 9,197,160 | -3.8% |
| **April** | 10,115,378 | 9,517,560 | -5.9% |
| **May** | 11,350,444 | 11,446,440 | +0.8% |
| **June** | 11,212,680 | 11,122,320 | -0.8% |
| **July** | 11,640,843 | 11,853,120 | +1.8% |
| **August** | 11,373,624 | 11,412,960 | +0.3% |
| **September** | 11,587,049 | 11,220,240 | -3.2% |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **94,424,336** | **91,575,840** | **-3.0%** |

## 3.2 Natural Gas Consumption

The actual natural gas consumption and cost in September 2021 decreased by **32.9%** and **32.6%** respectively compared to same month in 2019. To date, MGM Cotai has decreased actual natural gas consumption and associated costs by **25.0%** and **24.7%** respectively.

**Table 24. Cotai - Summary of Actual Monthly Natural Gas Consumption and Cost**

| Monthly Period | Actual Natural Gas Consumption ($m^3$) | | | Actual Natural Gas Cost (MOP) | | | % Change – Food Covers |
|---|---|---|---|---|---|---|---|
| | 2019 | 2021 | % Change - Actual | 2019 | 2021 | % Change - Actual Cost | |
| **January** | 112,881 | 60,643 | -46.3% | 671,460 | 363,990 | -45.8% | -53.9% |
| **February** | 99,721 | 54,221 | -45.6% | 594,320 | 325,930 | -45.2% | -46.8% |
| **March** | 112,506 | 68,066 | -39.5% | 669,261 | 407,980 | -39.0% | -42.2% |
| **April** | 107,194 | 86,866 | -19.0% | 638,120 | 518,960 | -18.7% | -42.5% |
| **May** | 103,138 | 90,102 | -12.6% | 614,347 | 537,940 | -12.4% | -36.0% |
| **June** | 87,535 | 80,606 | -7.9% | 522,890 | 482,270 | -7.8% | -52.7% |
| **July** | 82,825 | 102,473 | +23.7% | 495,276 | 610,450 | +23.3% | -38.7% |
| **August** | 79,387 | 51,837 | -34.7% | 475,084 | 311,790 | -34.4% | -62.5% |
| **September** | 76,717 | 51,460 | -32.9% | 459,259 | 309,560 | -32.6% | -51.1% |
| **October** | | | | | | | |
| **November** | | | | | | | |
| **December** | | | | | | | |
| **Total/Average** | **861,904** | **646,274** | **-25.0%** | **5,140,016** | **3,868,870** | **-24.7%** | **-47.5%** |

***Important note:*** *Minimum Outdoor Air Temperature (OAT is the main variable taken into consideration on natural gas consumption normalization.*

Energenz summarized the key items which could result in natural gas consumption decrease below:

- Boiler Natural Gas consumption decreased by 93.3% in September 2021 compared to same month in 2019. As the Heat Pump is used for maintaining DHW temperatures and disinfecting the hot water system instead of Natural Gas Boiler from 6 August 2021.
- Kitchen and Restaurants natural gas consumption, which constitutes to around 93.3% of the overall site consumption, decreased by 33.7% in September 2021 compared to same month in 2019, while food covers had decreased by 51.1%.

Table 25 presents the comparison of natural gas rates in the year of 2019 and 2021. The natural gas rate in September 2021 increased by **0.5%** compared with the same month in 2019.

**Table 25. Cotai - Summary of Actual Monthly Natural Gas Rates**

| Monthly Period | Year 2019 – Natural Gas Rate (MOP/m³) | Year 2021 – Natural Gas Rate (MOP/m³) | % Change – Natural Gas Rate |
|---|---|---|---|
| **January** | 5.95 | 6.00 | +0.9% |
| **February** | 5.96 | 6.01 | +0.9% |
| **March** | 5.95 | 5.99 | +0.8% |
| **April** | 5.95 | 5.97 | +0.4% |
| **May** | 5.96 | 5.97 | +0.2% |
| **June** | 5.97 | 5.98 | +0.2% |
| **July** | 5.98 | 5.96 | -0.4% |
| **August** | 5.98 | 6.01 | +0.5% |
| **September** | 5.99 | 6.02 | +0.5% |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **5.97** | **5.99** | **+0.4%** |

## 3.3 ISO50001 Energy Consumption Record

Cotai ISO50001 energy target is to achieve 13% normalized energy consumption reduction by 2025 compared to 2019 baseline. Based on the official agreement with MGM Cotai, e-bus consumption should be included for the energy consumption tracking as part of the ISO50001 requirement. Table 26 below summarizes the site monthly energy consumption (Electricity and natural gas combined) compared to 2019 baseline. The year-to-date normalized energy saving achieved in September 2021 is **6.1%.**

**Table 26. Cotai ISO 50001 Normalized Monthly Energy Consumption**

| Monthly Period | 2019 Normalized Energy Consumption (GJ) | 2021 Actual Energy Consumption (GJ) | % Change - Normalized (2021 Vs 2019) |
|---|---|---|---|
| **January** | 35,529 | 31,511 | -11.3% |
| **February** | 36,539 | 29,377 | -19.6% |
| **March** | 38,505 | 35,472 | -7.9% |
| **April** | 40,368 | 37,277 | -7.7% |
| **May** | 44,558 | 44,334 | -0.5% |
| **June** | 44,002 | 42,837 | -2.6% |
| **July** | 45,495 | 46,227 | +1.6% |
| **August** | 44,554 | 42,885 | -3.7% |
| **September** | 45,225 | 42,179 | -6.7% |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **374,774** | **352,099** | **-6.1%** |

***Important note**: Factors used for fuels' unit conversion: 1) Electricity (0.0036) and Natural Gas (0.0347), e-bus consumption is included*

## 3.4 Water Consumption

According to Table 27, the actual water consumption in September 2021 decreased by **2.6%** compared to same month in 2019. To date, MGM Cotai has decreased actual water consumption and associated costs by **14.2%** and **13.9%** respectively.

**Table 27. Cotai - Summary of Actual Monthly Water Consumption and Cost**

| Monthly Period | Actual Water Consumption ($m^3$) | | | Actual Water Cost (MOP) | | | Normalization Factors | |
|---|---|---|---|---|---|---|---|---|
| | 2019 | 2021 | % Change Actual | 2019 | 2021 | % Change Actual Cost | % Change – CDD | % Change – Food Covers |
| **January** | 57,331 | 43,295 | -24.5% | 445,450 | 336,678 | -24.4% | -18.5% | -53.9% |
| **February** | 65,011 | 50,847 | -21.8% | 505,050 | 395,188 | -21.8% | -3.8% | -46.8% |
| **March** | 63,136 | 55,680 | -11.8% | 488,590 | 432,418 | -11.5% | +39.0% | -42.2% |
| **April** | 73,577 | 65,523 | -10.9% | 571,320 | 509,018 | -10.9% | -5.5% | -42.5% |
| **May** | 73,317 | 77,517 | +5.7% | 569,350 | 612,228 | +7.5% | +43.8% | -36.0% |
| **June** | 79,312 | 66,127 | -16.6% | 615,790 | 513,528 | -16.6% | -4.4% | -52.7% |
| **July** | 86,836 | 70,726 | -18.6% | 674,210 | 549,158 | -18.5% | +1.4% | -38.7% |
| **August** | 85,257 | 62,748 | -26.4% | 661,940 | 487,412 | -26.4% | -4.7% | -62.5% |
| **September** | 72,406 | 70,538 | -2.6% | 562,250 | 547,722 | -2.6% | +7.9% | -51.1% |
| **October** | | | | | | | | |
| **November** | | | | | | | | |
| **December** | | | | | | | | |
| **Total/ Average** | **656,183** | **563,001** | **-14.2%** | **5,093,950** | **4,383,348** | **-13.9%** | **+5.9%** | **-47.5%** |

***Important note**: Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

Energenz summarized the key items which could result in water consumption decrease below:

- Hotel Tower & Podium water consumption decreased by 29.3% in September 2021 compared to same month in 2019. Moreover, guest occupancy decreased 54.7% in September 2021 compared to same month in 2019.

Table 28 presents the comparison of water rates in the year of 2019 and 2021. The water rate in September 2021 slightly decreased by **0.004%** compared to same month in 2019. The average utility rate of water has increased by **0.3%** compared to 2019.

**Table 28. Cotai - Summary of Actual Monthly Water Rates**

| Monthly Period | Year 2019 – Water Rate (MOP/$m^3$) | Year 2021 – Water Rate (MOP/$m^3$) | % Change – Water Rate |
|---|---|---|---|
| **January** | 7.77 | 7.78 | +0.1% |
| **February** | 7.77 | 7.77 | +0.04% |
| **March** | 7.74 | 7.77 | +0.4% |
| **April** | 7.76 | 7.77 | +0.05% |
| **May** | 7.77 | 7.90 | +1.7% |
| **June** | 7.76 | 7.77 | +0.02% |
| **July** | 7.76 | 7.76 | +0.01% |
| **August** | 7.76 | 7.77 | +0.05% |
| **September** | 7.77 | 7.76 | -0.004% |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **7.76** | **7.78** | **+0.3%** |

Table 29 summarized the monthly normalized water consumption. The normalized water consumption in September 2021 increased by **3.7%** compared to the actual consumption in 2021.

**Table 29. Cotai - Summary of Normalized Monthly Water Consumption**

| Monthly Period | 2019 Normalized Water Consumption ($m^3$) | 2021 Actual Water Consumption ($m^3$) | % Change - Normalized |
|---|---|---|---|
| **January** | 45,293 | 43,295 | -4.4% |
| **February** | 48,765 | 50,847 | +4.3% |
| **March** | 55,357 | 55,680 | +0.6% |
| **April** | 58,684 | 65,523 | +11.7% |
| **May** | 69,926 | 77,517 | +10.9% |
| **June** | 63,635 | 66,127 | +3.9% |
| **July** | 71,521 | 70,726 | -1.1% |
| **August** | 63,252 | 62,748 | -0.8% |
| **September** | 66,339 | 70,538 | +6.3% |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **542,771** | **563,001** | **+3.7%** |

## 3.5 Cotai Year 2021 Projects

Energenz will perform analysis and coordinate with MGM to implement the following projects in 2021 and reduce the site energy and water consumption which supporting Energenz annual normalized energy savings target of 2%.

**Table 30. MGM Cotai Year 2021 Projects**

| Item | Initiatives |
|---|---|
| **Electricity** | |
| 1 | Air Change Rate Optimization |
| 2 | B1 Carpark and Loading Bay Lighting Operation Schedule Optimization |
| 3 | Kitchen Exhaust Fan (KEF) Demand Control Ventilation (DCV) and Hood Control for Remaining Kitchen's (e.g. Banquet, Chun, Mansion, Coast.) |
| 4 | Optimize BOH FCU and VAV Temperature Set-Point |
| 5 | Optimize Car Park Ventilation Fans Control |
| 6 | Optimize Hotel Tower Circulation Pumps Operation |
| 7 | Chiller Plant Optimization:<br>1. Review Chillers Staging<br>2. Cooling Tower Fan Speed Optimization<br>3. Condensing Water Pump Speed Optimization |
| 8 | Chilled Water Differential Pressure Set-Point Reset Logic Implementation |
| 9 | B2 Domestic Hot Water Heat Exchanger Upgrade for fully utilizing the heat pump |
| 10 | Implement Static Pressure & Supply Air Temperature Set-Point Reset in AHU (48 Units) |
| 11 | PICCV Implementation (7 Units) |
| 12 | Solar PV Panel Implementation |
| **Water** | |
| 13 | Extension of CDP from ST-CD-02 to ST-CD-01 for Cooling Tower Makeup |

# Appendix A

## Regression Analysis

The 2019 normalized consumption modelling is remained used for tracking the agreed normalized energy savings target of 4% accurately for MGM Macau and MGM Cotai. The following tables show a summary of the selected independent variables for each utility which are used for the normalized consumption modelling in 2021.

### MGM Macau

| Utilities | 2021 | |
|---|---|---|
| **Electricity (kWh)** | CDD (Base T=17°C) | |
| **LPG (kg)** | Min OAT | |
| **Water ($m^3$)** | CDD (Base T=17°C) | Food Cover |

### MGM Cotai

| Utilities | 2021 | |
|---|---|---|
| **Electricity (kWh)** | CDD (Base T=17°C) | |
| **Natural Gas ($m^3$)** | Min OAT | |
| **Water ($m^3$)** | CDD (Base T=17°C) | Food Cover |

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